Integer Linear Programming
Algorithmic Methods for Mathematical Models

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Problem 1: Food Production Optimization

Problem Statement

A food is manufactured by refining raw oils and blending them together. The raw oils come in two categories:
Vegetable oil: VEG 1, VEG 2
Non-vegetable oil: NON 1, NON 2, NON 3
Vegetable oils and non-vegetable oils require different production lines for refining. It is not possible to refine more than 200 tons of vegetable oil and more than 250 tons of non-vegetable oils. There is no loss of weight in the refining process and the cost of refining may be ignored.
There is a technological restriction of hardness in the final product. In the units in which hardness is measured, this must lie between 3 and 6. We assume the hardness of final product is the weighted average of hardness of the raw oils.
Data:

	VEG 1	VEG 2	NON 1	NON 2	NON 3
Hardness	8.8	6.1	2.0	4.2	5.0
Cost (€/ton)	110	120	130	110	115

The final product sells at 150€ per ton.
Additional conditions:

- The food should be made up of at least three oils.
- If an oil is used then at least 20 tons must be used.
- If either of VEG 1 or VEG 2 is used then NON 3 must also be used.

Objective: What manufacturing policy should the company pursue in order to maximize profit? Formulate the problem as an MIP.

Part (a): Basic LP Formulation

Decision variables:

$$TVEG_1, TVEG_2 = \text{tons of vegetable oils VEG1, VEG2 used} \tag{1}$$
$$TNON_1, TNON_2, TNON_3 = \text{tons of non-vegetable oils NON1, NON2, NON3 used} \tag{2}$$
$$PROD = TVEG_1 + TVEG_2 + TNON_1 + TNON_2 + TNON_3 \text{ (total production)} \tag{3}$$

Objective function:

$$\text{Maximize } Z = 150 \cdot PROD - 110 \cdot TVEG_1 - 120 \cdot TVEG_2 \tag{4}$$
$$- 130 \cdot TNON_1 - 110 \cdot TNON_2 - 115 \cdot TNON_3 \tag{5}$$

Basic constraints:

Vegetable oil refining capacity: $TVEG_1 + TVEG_2 \leq 200$ (6)

Non-vegetable oil refining capacity: $TNON_1 + TNON_2 + TNON_3 \leq 250$ (7)

Minimum hardness: (8)

$$\frac{8.8 \cdot TVEG_1 + 6.1 \cdot TVEG_2 + 2.0 \cdot TNON_1 + 4.2 \cdot TNON_2 + 5.0 \cdot TNON_3}{PROD} \geq 3 \tag{9}$$

Maximum hardness: (10)

$$\frac{8.8 \cdot TVEG_1 + 6.1 \cdot TVEG_2 + 2.0 \cdot TNON_1 + 4.2 \cdot TNON_2 + 5.0 \cdot TNON_3}{PROD} \leq 6 \tag{11}$$

Total production definition: (12)

$$PROD = TVEG_1 + TVEG_2 + TNON_1 + TNON_2 + TNON_3 \tag{13}$$

$$TVEG_1, TVEG_2, TNON_1, TNON_2, TNON_3 \geq 0 \tag{14}$$

Part (b): MIP Formulation with Additional Conditions

Additional binary variables:

$$UVEG_1, UVEG_2 = \begin{cases} 1 & \text{if vegetable oil VEG1, VEG2 is used} \\ 0 & \text{otherwise} \end{cases} \tag{15}$$

$$UNON_1, UNON_2, UNON_3 = \begin{cases} 1 & \text{if non-vegetable oil NON1, NON2, NON3 is used} \\ 0 & \text{otherwise} \end{cases} \tag{16}$$

Additional constraints for extra conditions:

$$\text{At least 3 oils: } UVEG_1 + UVEG_2 + UNON_1 + UNON_2 + UNON_3 \geq 3 \tag{17}$$

$$\text{Minimum usage per oil:} \tag{18}$$

$$TVEG_1 \geq 20 \cdot UVEG_1, \quad TVEG_2 \geq 20 \cdot UVEG_2 \tag{19}$$

$$TNON_1 \geq 20 \cdot UNON_1, \quad TNON_2 \geq 20 \cdot UNON_2, \quad TNON_3 \geq 20 \cdot UNON_3 \tag{20}$$

$$\text{Maximum usage per oil:} \tag{21}$$

$$TVEG_1 \leq M \cdot UVEG_1, \quad TVEG_2 \leq M \cdot UVEG_2 \tag{22}$$

$$TNON_1 \leq M \cdot UNON_1, \quad TNON_2 \leq M \cdot UNON_2, \quad TNON_3 \leq M \cdot UNON_3 \tag{23}$$

$$\text{VEG}\rightarrow\text{NON3 implication: } UVEG_1 + UVEG_2 \leq 2 \cdot UNON_3 \tag{24}$$

$$UVEG_1, UVEG_2, UNON_1, UNON_2, UNON_3 \in \{0, 1\} \tag{25}$$

Binary Variable Transformations

How to find the transformations for binary variables?
Let b be a binary variable and $f(x)$ a linear expression.
Case 1: We want to express with linear inequalities that "if $b = 0$ then $f(x) \leq 0$":
Let U be an upper bound for the value of $f(x)$ over all feasible solutions. Then, for any feasible solution, the following constraint holds:

$$f(x) \leq U \cdot b$$

If $b = 0$, then $f(x) \leq 0$. If $b = 1$, then $f(x) \leq U$, which is always true.
Case 2: We want to express with linear inequalities that "if $b = 1$ then $f(x) \geq 0$":
Let L be a lower bound for the value of $f(x)$ over all feasible solutions. Then, for any feasible solution, the following constraint holds:

$$f(x) \geq L \cdot (1 - b)$$

If $b = 1$, then $f(x) \geq 0$. If $b = 0$, then $f(x) \geq L$, which is always true.

In case we have "if $b = 0$ then $f(x) \geq k$": We can transform our function by defining a new function $f'(x) = f(x) - k$. Then, we can apply the same logic as in Case 2: Let L' be a lower bound for the value of $f'(x)$ over all feasible solutions. Then, for any feasible solution, the following constraint holds:

$$f'(x) \geq L' \cdot (1 - b)$$

Which can be rewritten as:

$$f(x) - k \geq L' \cdot (1 - b)$$

or equivalently:

$$f(x) \geq k + L' \cdot (1 - b)$$

Problem 2: Sudoku

Given the following Sudoku puzzle,

5	3			7				
6			1	9	5			
	9	8					6	
8				6				3
4			8		3			1
7				2				6
	6					2	8	
			4	1	9			5
				8			7	9

formulate the problem of completing it as a MIP.

MIP Formulation

Decision variables:

- x_{ijk}: binary variable that is 1 if the cell in row i and column j contains the number k, and 0 otherwise.
  Where $i, j, k \in \{1, 2, \ldots, 9\}$.

Objective function: There is no objective function. To cost the problem as a MIP, we can take arbitrarily any function, for example: we can minimize 0.

Constraints:

1. At each row, each value appears exactly once:

$$\sum_{i=1}^{9} x_{ijk} = 1 \quad \forall j, k \in \{1, \ldots, 9\}$$

2. At each column, each value appears exactly once:

$$\sum_{j=1}^{9} x_{ijk} = 1 \quad \forall i, k \in \{1, \ldots, 9\}$$

3. In each 3x3 subsquare, each value appears exactly once:

   Let's us identify each of the subsquares with the set of coordinates if its cells:

   - $\mathbf{S_{11}} = \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)\}$

$$\begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{bmatrix}$$

   For each subsquare S, for all values k:

$$\sum_{(i,j) \in S} x_{ijk} = 1 \quad \forall k \in \{1, \ldots, 9\}$$

4. Fixed cells are respected.
   For each fixed cell (i,j) with value k:

$$x_{ijk} = 1$$

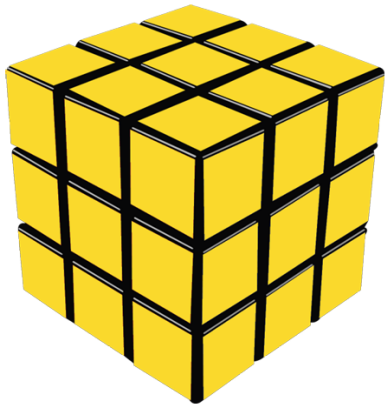
5. At each cell (i,j), there is exactly one value:

$$\sum_{k=1}^{9} x_{ijk} = 1 \quad \forall i, j \in \{1, \ldots, 9\}$$

Problem 3: Twenty-Seven Cells Arrangement

Problem Statement

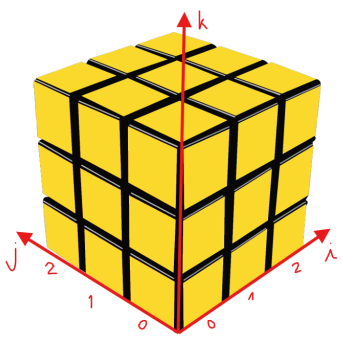
Twenty-seven cells are arranged in a three-dimensional $3 \times 3 \times 3$ cube as shown below.



Three cells are regarded as lying in the same line if they are on the same horizontal or vertical line or the same diagonal. Diagonals exist on each horizontal and vertical section and connecting opposite vertices of the cube. (There are 49 lines altogether.)
Given 13 white balls and 14 black balls, arrange them, one to a cell, so as to minimise the number of lines with balls all of one colour.

MIP Formulation

Let's identify each of the three cells in the cube with a triplet (i, j, k) where $i, j, k \in \{0, 1, 2\}$.
Moreover, let's identify a line with the set of coordinates of its cells.



Decision variables:

- x_{ijk} = the ball at cell (i, j, k) is white (1) or black (0).
  Where $i, j, k \in \{0, 1, 2\}$. (of type binary)
- y_l = the ball in line l are all of the same color (1) or not (0).
  Where $l \in \{1, 2, \ldots, 49\}$ (the 49 lines in total). (of type binary)

Objective function: Minimize the number of lines with balls all of one colour:

$$\text{minimize} \sum_{l=1}^{49} y_l$$

Constraints:

1. There are 13 white balls and 14 black balls:

$$\boxed{\sum_{(i,j,k)\in\text{cells}} x_{ijk} = 13} \quad \text{and} \quad \boxed{\sum_{(i,j,k)\in\text{cells}} (1 - x_{ijk}) = 14} \tag{26}$$

2. If all balls in a line l have the same colour, then $y_l = 1$.
   Let l be the line consisting of the cells $(i_0, j_0, k_0), (i_1, j_1, k_1), (i_2, j_2, k_2)$. Let's consider two cases:

   (a) If all balls in line l are **white** , then $y_l = 1$:

$$x_{i_0j_0k_0} \wedge x_{i_1j_1k_1} \wedge x_{i_2j_2k_2} \Rightarrow y_l$$

$$\neg(x_{i_0j_0k_0} \wedge x_{i_1j_1k_1} \wedge x_{i_2j_2k_2}) \vee y_l \quad \text{(by implication law)}$$

   If $a_1, a_2, \ldots, a_n, b_1, b_2, \ldots, b_m$ are binary variables, then:

$$\neg(a_1 \wedge a_2 \wedge \cdots \wedge a_n) \vee (b_1 \wedge b_2 \wedge \cdots \wedge b_m) \Leftrightarrow \sum_{i=1}^{n}(1 - a_i) + \sum_{j=1}^{m} b_j \geq 1$$

   Therefore, we can express our constraint as:

$$\boxed{(1 - x_{i_0j_0k_0}) + (1 - x_{i_1j_1k_1}) + (1 - x_{i_2j_2k_2}) + y_l \geq 1}$$

   (b) If all balls in line l are **black** , then $y_l = 1$:

$$(1 - x_{i_0j_0k_0}) \wedge (1 - x_{i_1j_1k_1}) \wedge (1 - x_{i_2j_2k_2}) \Rightarrow y_l$$

$$\neg((1 - x_{i_0j_0k_0}) \wedge (1 - x_{i_1j_1k_1}) \wedge (1 - x_{i_2j_2k_2})) \vee y_l$$

   Applying the same reasoning as in the previous case, we can express our constraint as:

$$\boxed{x_{i_0j_0k_0} + x_{i_1j_1k_1} + x_{i_2j_2k_2} + y_l \geq 1}$$

Problem 4: Infectious Disease Outbreak Investigation

Problem Statement

An outbreak of an infectious disease has been observed in a set N of locations. There exists a set M of teams capable of investigating these outbreaks. Team $i \in M$ can conclude its investigation of the outbreak in location $j \in N$ in t_{ij} hours. Each team can either investigate zero, one, or two outbreaks. If a team investigates two outbreaks, they must travel from one location to the next. The travel time from location $j_1 \in N$ to location $j_2 \in N$ is $d_{j_1 j_2}$. Travel times can be assumed to be symmetric. Once all outbreaks have been investigated, a disease control center can take action to combat the outbreak. The goal is to minimize the amount of time necessary to complete the investigation of all locations.

MIP Formulation

Decision variables:

- $x_{ij} = 1$ if team i investigates location j, 0 otherwise.
  Where $1 \leq i \leq M$ and $1 \leq j \leq N$. (of type binary)
- $y_{ij_1j_2} = 1$ if team i investigates locations j_1 and j_2 in that order, 0 otherwise.
  Where $1 \leq i \leq M$ and $1 \leq j_1, j_2 \leq N$. (of type binary)
- z = the maximum time needed for any team to complete its investigations. (of type real non negative)

Constraints:

1. Each team can investigate at most two locations:

$$\boxed{\sum_{j=1}^{N} x_{ij} \leq 2 \quad \forall 1 \leq i \leq M}$$

2. Each location must be investigated by exactly one team:

$$\boxed{\sum_{i=1}^{M} x_{ij} = 1 \quad \forall 1 \leq j \leq N}$$

3. Variables are linked: $y_{ij_1j_2} \iff (x_{ij_1} \wedge x_{ij_2})$:

   (a) $y_{ij_1j_2} \implies (x_{ij_1} \wedge x_{ij_2})$:

$$\neg y_{ij_1j_2} \vee (x_{ij_1} \wedge x_{ij_2})$$

$$(\neg y_{ij_1j_2} \vee x_{ij_1}) \wedge (\neg y_{ij_1j_2} \vee x_{ij_2})$$

$$(1 - y_{ij_1j_2}) + x_{ij_1} \geq 1 \text{ and } (1 - y_{ij_1j_2}) + x_{ij_2} \geq 1$$

   Therefore, we can express our constraints as:

$$\boxed{y_{ij_1j_2} \leq x_{ij_1}} \quad \text{and} \quad \boxed{y_{ij_1j_2} \leq x_{ij_2}} \quad \forall 1 \leq i \leq M, 1 \leq j_1, j_2 \leq N$$

(b) $y_{ij_{1}j_{2}} \impliedby (x_{ij_1} \wedge x_{ij_2})$:

$$(x_{ij_1} \wedge x_{ij_2}) \implies y_{ij_1j_2}$$
$$\neg(x_{ij_1} \wedge x_{ij_2}) \vee y_{ij_1j_2}$$
$$\neg x_{ij_1} \vee \neg x_{ij_2} \vee y_{ij_1j_2}$$
$$(1 - x_{ij_1}) + (1 - x_{ij_2}) + y_{ij_1j_2} \geq 1$$

Therefore, we can express our constraint as:

$$\boxed{x_{ij_1} + x_{ij_2} - y_{ij_1j_2} \leq 1} \quad \forall 1 \leq i \leq M, 1 \leq j_1, j_2 \leq N$$

4. z has its intended meaning: For every team i, the time needed for team i to complete its investigations must be less than or equal to z:

$$\boxed{\sum_{j=1}^{N} t_{ij}x_{ij} + \sum_{j_1,j_2=1;j_1<j_2}^{N} d_{j_1j_2} \cdot y_{ij_1j_2} \leq z \quad \forall 1 \leq i \leq M}$$

Objective function: Minimize the time needed to complete the investigation of all locations.

$$\min \max_{1 \leq i \leq M} \{\text{time needed for team } i\}$$

The time needed for team i to complete its investigations is

$$\sum_{j=1}^{N} t_{ij}x_{ij} + \sum_{j_1=1}^{N}\sum_{j_2=1}^{N} d_{j_1j_2}(x_{ij_1} \wedge x_{ij_2}) \implies \sum_{j_1,j_2=1;j_1<j_2}^{N} d_{j_1j_2} \cdot y_{ij_1j_2}$$

Problem 5: Graph Coloring

Problem Statement

Consider the problem of graph coloring: given a graph $G = (V, E)$, we want to find the minimum number of colors to paint the vertices so that adjacent vertices have different colors. Model this problem as a MIP.

MIP Formulation

Let's label vertices $1, 2, \ldots, n = |V|$.

Decision variables:

- c_v = color assigned to vertex v (for all $v \in V$) (of type integer non negative)
- z = intuitively, $\max_{v \in V} c_v$ (of type integer non negative)
- $z_{uv} = 1$ if $c_u < c_v$, 0 if $c_u > c_v$ (for all $(u, v) \in E$) (of type binary)

Objective function: Minimize the number of colors used:

$$\min z$$

Constraints:

1. z 'behaves' as $\max_{v \in V} c_v$:

$$\boxed{c_v \leq z \quad \forall v \in V}$$

2. Adjacent vertices have different colors:

$$\boxed{c_u \neq c_v \quad \forall (u,v) \in E}$$

The problem is that this equation is not linear. We can transform it using binary variables:

This is the same as saying that either $c_u < c_v$ or $c_u > c_v \implies c_u + 1 \leq c_v \vee c_u \geq c_v + 1$

We define a new variable z_{uv} such that:

$$z_{uv} = \begin{cases} 1 & \text{if } c_u < c_v \\ 0 & \text{if } c_u > c_v \end{cases}$$

We can express our constraint as:

$$z_{uv} = 1 \implies c_u + 1 \leq c_v \implies c_v - c_u - 1 \geq 0$$

$$z_{uv} = 0 \implies c_u \geq c_v + 1 \implies c_u - c_v - 1 \geq 0$$

Applying the transformations for binary variables explained in Problem 1, we get:

If $b = 0 \longrightarrow f(x) \leq 0$ is equivalent to $f(x) \leq U \cdot b$, where U is an upper bound for $f(x)$.

If $b = 1 \longrightarrow f(x) \geq 0$ is equivalent to $f(x) \geq L \cdot (1 - b)$, where L is a lower bound for $f(x)$.

- Let's express $z_{uv} = 1 \implies c_v - c_u - 1 \geq 0$ We know that $c_v \geq 1 \implies c_u \leq n \implies -c_u \geq -n$
  Therefore, a lower bound is $L = -n$.
  So, the implication is equivalent to:

$$c_v - c_u - 1 \geq -n \cdot (1 - z_{uv})$$

- Let's express $z_{uv} = 0 \implies c_u - c_v - 1 \geq 0$ We know that $c_v \leq n \implies c_u \geq 1 \implies -c_u \leq -1$
  Therefore, an upper bound is $U = n$.
  So, the implication is equivalent to:

$$c_u - c_v - 1 \leq n \cdot z_{uv}$$

Therefore, we can express our constraints as:

$$\boxed{c_v - c_u - 1 \geq -n \cdot (1 - z_{uv})} \quad \text{and} \quad \boxed{c_u - c_v - 1 \leq n \cdot z_{uv}} \quad \forall (u,v) \in E$$

Alternative MIP Formulation

Another possibility of defining the decision variables is:

- $x_{vc} = 1$ if vertex v is assigned color c, 0 otherwise (for all $v \in V$ and $1 \leq c \leq n$) (of type binary)
- $u_k = 1$ if color k is used, 0 otherwise (for all $1 \leq k \leq n$) (of type binary)
- z = number of colors used (of type integer non negative)

Objective function: Minimize the number of colors used:

$$\min z$$

Constraints:

1. Variables are related: For each $1 \leq v \leq n$ and $1 \leq k \leq n$:

$$x_{vk} \to u_k \quad \Longrightarrow \quad \neg x_{vk} \vee u_k \quad \Longrightarrow \quad (1 - x_{vk}) + u_k \geq 1 \quad \Longrightarrow \quad \boxed{u_k \geq x_{vk}}$$

2. Neighbors are painted with different colors: For each edge $(u, v) \in E$ and $1 \leq k \leq n$:

$$\neg(x_{uk} \wedge x_{vk}) \quad \Longrightarrow \quad (1 - x_{uk}) + (1 - x_{vk}) \geq 1 \quad \Longrightarrow \quad \boxed{x_{uk} + x_{vk} \leq 1}$$

3. Each vertex is painted with exactly one color: For each $1 \leq v \leq n$:

$$\boxed{\sum_{k=1}^{n} x_{vk} = 1}$$

   Observation: This constraint was not in our first formulation because it was implicit in the definition of the variables c_v.